Simply Produced Closed-Head, Focal Lesions for Study of Repair in Immature Rat Cerebral Cortex

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Geller, Lester M., Cowen, David, and Wolf, Abner (1970). *Simply Produced Closed-Head, Focal Lesions for Study of Repair in Immature Rat Cerebral Cortex.* DEVELOPMENTAL PSYCHOBIOLOGY, 3(4): 217-223. Multiple electroconvulsive shocks consistently and rapidly produce uniform, bilaterally symmetrical lesions in the immature rat's cerebral cortex, without the incidental damage that accompanies other experimental injuries to the central nervous system. The terminal lesion, devoid of nerve cells, is a narrow, laminar band of numerous myelinated axons, unaccompanied by a glial scar. Insofar as the initially damaged area becomes occupied by nerve fibers, the healing process may be thought of as involving a regeneration. A variety of studies for which the electroshock lesions might be used is suggested.

closed-head, focal cerebral cortical lesions laminar cortical lesions
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Repeated application of electric current to the infant rat's head by means of externally placed electrodes produces small, bilateral, symmetrical lesions primarily in layers iv and v of the cerebral cortex (Fig. 1A and B), useful for the study of healing processes in immature neural tissue. This method of making a lesion avoids immediate damage to the blood vessels and leptomeninges, gross hemorrhage, and total tissue necrosis, which complicate the study of wound repair in the central nervous system after injury produced by other means (e.g., excision, laceration, puncture, freezing, and ultrasonic, laser beam, or high-dose ionizing radiation). Additional advantages of the method are the simplicity of the procedure and of the apparatus required, the almost 100% survival rate after treatment (although a death may occasionally occur over the next few days), and the avoidance of surgical incision of the scalp and skull. Indeed, so far as we know, no other method of equal simplicity exists for the production of similar lesions. Much the same type of laminar cortical lesion has, however, been produced in older animals with a beam of heavy ionizing particles, but this requires the use of a cyclotron or linear accelerator (Estable-Puig, de Estable, Tobias, & Haymaker, 1964; Janssen, Klatzo, Miquel, Brustad, Behar, Haymaker, Lyman, Henry, & Tobias, 1962; Kruger & Clemente, 1964; Malis, Baker, Kruger, & Rose, 1957; Malis, Loevinger, Kruger, & Rose, 1957; Maxwell & Kruger, 1964; Rose, Malis, Kruger, & Baker, 1960).

The electrodes used in our procedure are made from silver wire, balled at one end (diameter, 2.5-3mm), and coiled to make them slightly springy. The balled end is wrapped in layers of cotton-filled gauze sponge, tied down with surgical silk thread, and neatly trimmed beyond the tie, to yield a fairly soft or spongy, rounded surface. The free end of each electrode is mounted in a pair of plastic pliers so that the interelectrode distance can be easily adjusted for differences in skull size among infant rats. Leads soldered to the mounted electrodes are plugged into a 60 cycle AC constant voltage stimulator, made simply from a Variac (0-120 V range) connected in series with a voltage regulating transformer. In series with the stimulator is an electronic timer (General Electric, model no. CR7504-A142), calibrated in 0.05 sec steps, with a timing range of 0 to 1.05 sec. The timer, operated by a foot switch, permits current to flow from the stimulator to the electrodes for a preset interval only.

Before use, the electrodes are dipped in physiological saline which, for our purpose, provides a considerably better, less irritating medium for electrode-skin contact than electrode paste. While the electrodes and their site of placement should be well moistened with saline, care should be taken to avoid wetting neighboring skin areas lest extraneous current paths be established. With the animal lightly held on a wooden table by the experimenter's thumb and forefinger, the electrodes are firmly applied to either side of the head, just medial and rostral to the base of the ear buds, and the current is turned on for 0.3 sec. The voltage used is 85 V. The animal responds to the electrostimulation with a

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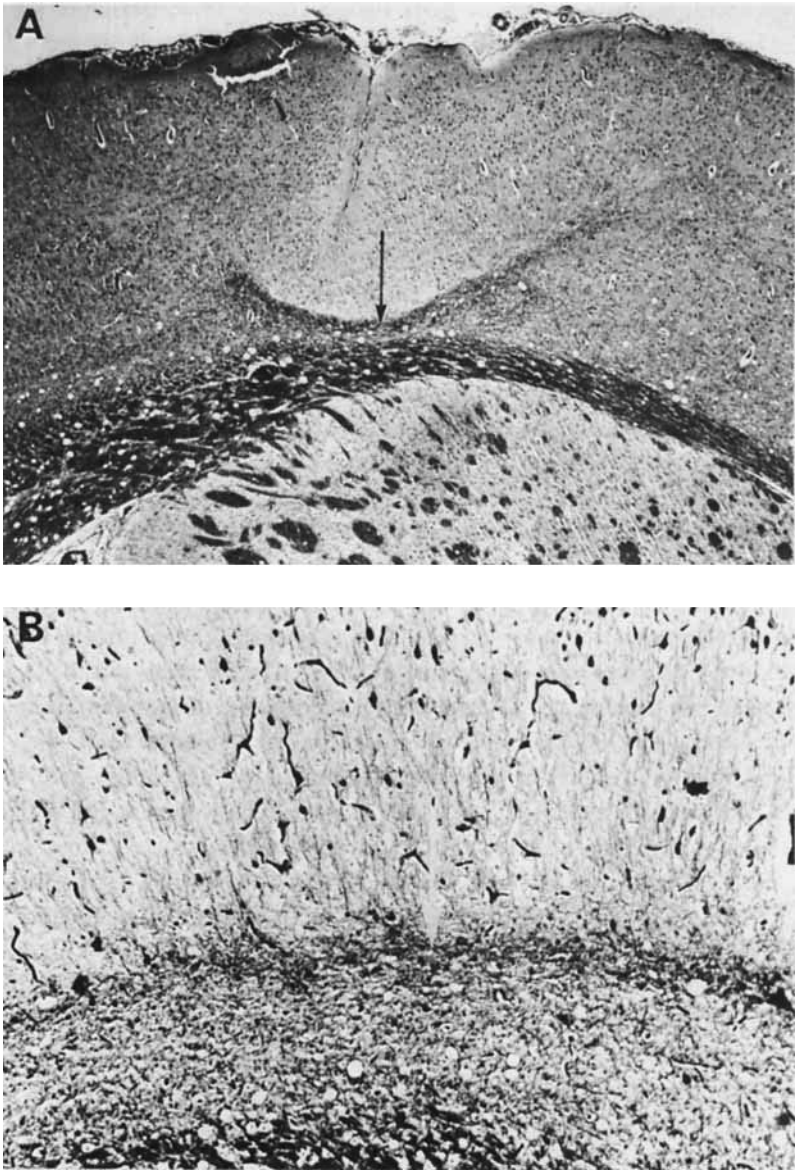


Fig. 1. (A) Brain of rat 18 months after the electroshock procedure. Lesion (arrow) is a narrow band of hypermyelination in lamina v, curving outward. The cortex just above the band is pale due to decreased myelination. The normal cerebral white matter (broad, black bundle) and basal ganglia are seen below. Luxol Fast blue-cresyl violet stain (X 32). (B) Laminar band of abnormally heavy myelination (arrow), consisting of a superabundance of myelinated axons, within layer v of cortex of rat 16 months after electroshock. Beneath it the cortex and a small segment of subjacent white matter are normally myelinated. Superficial to it radially-oriented myelin sheaths are reduced in number; dark black structures here are incidentally-stained capillaries and some nerve cells. Mahon stain (X 130).

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tonic convulsion characterized by body rigidity; severe tonic flexion of the rear limbs and often forward extension of the front limbs, the latter sometimes displaying delicate fluttering activity; downward movement of the forelimbs so that they extend rigidly below and toward the rear of the animal; rising, outward arching of the back and expansion of the chest, often a brief cessation of respiration with accompanying cyanosis; and digging and/or fluttering movements of the limbs. Thirty shocks are administered in a 3-4 min interval. This period provides ample time for relaxation from the rigidity induced by the preceding shock and, in most cases, for return of the animal's color. It is also long enough for as many as 8 rats, even more with experience, to be treated, one after the other, before the turn of the first-shocked rat occurs again. During the 90-120 min required for the entire procedure the litter is kept away from the mother and warmed under a 60-watt bulb. Control animals from each litter are subjected to the same manipulations, i.e., 30 applications of electrodes to the head, at a 3-4 min interval, but without the electrostimulation.

The lesion-producing factors used (0.3 sec stimulation, 85 V, 30 shocks, 3 to 4 min interval between shocks) were chosen after the effects of many different combinations had been studied. The factors selected regularly lead to the rapid development of a uniform series of pathological changes in the cerebral cortex. In good part the minimal damage produced in the scalp, and the remarkably small loss of animals, can also be attributed to the factors used. (Some of our results suggest, however, that 30 shocks may be excessive; quite likely the minimal number required to produce a lesion, keeping the other factors constant, is between 20 and 30.) The lesions are located in the line of, and subjacent to, the site of electrode placement. Presumably they are mainly the result of electrically generated heat, as are small foci of injury in the skin at the electrode sites. Other factors involved in the passage of an alternating current through brain tissue, e.g., ionic and molecular oscillations (Zeman, 1968), may contribute to the damage, as may additional, though poorly defined, so-called specific electric effects. The latter, Zeman (1968) indicates, have been invoked by some workers to account for histopathological changes induced in the brain by currents supposedly too weak to have yielded significant thermal effects.

In our study of the reaction of the developing cerebral cortex to injury, Columbia-Sherman rats (*Rattus norvegicus*), 4 to 4 1/2 days old when treated, were killed immediately at the end of the 30 shocks, at 3-hr intervals for the next 24 hr, then at 36 and 48 hr, thereafter daily for the next 5 days, then at a 1 to 7 day interval for the following 9 weeks, and later at more widely spaced periods until 19 months of age. Histological examination of the brains revealed no lesions in the control animals. The severe seizures associated with the shocks did not lead to any apparent neuropathological changes such as might have been anticipated as a result of repeated short episodes of hypoxia. Briefly, the development and resolution of the lesions included the rapid occurrence of a localized

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focus of edema and a loss of nerve cells, a short period of phagocytosis, a proliferation of myelin-forming glial cells, and the appearance of an abundance of myelinated axons in the damaged area. A detailed account of the pathological changes and a discussion of the histopathogenesis of these lesions has recently been published (Cowen, Geller, & Wolf, 1970). The lesion in its final form was a clearly demarcated, narrow, laminar band of myelinated fibers exceeding in number that of the normal cortex (Fig. 1A and B). It was free of glial scarring. Superficial to the lesion myelination of the cortex was often deficient. Although the final, laminar lesion did not contain nerve cell bodies, the deeper portion of the original, considerably larger, zone of injury had been repopulated by them, but their origin, at this point, is unclear.

The occupation of the damaged area by axons, which became myelinated, may be thought of as constituting a regeneration. Whether this resulted from a regrowth of fibers from the proximal segments of initially injured axons, or from the entry into the lesion of fibers normally growing in the cortex during its active development, or perhaps from both, has yet to be determined. We believe that with appropriate techniques (electron microscopy, autoradiography, histochemistry) the electroshock lesions can be profitably employed for the study of such questions, as well as of other interesting and important topics, for example, differentiation of glial and nerve cell precursors and of glial cells *per se*, development of axons and of myelin sheaths, formation of abnormal glia, cell migration, and the regenerative capacity of nerve cells, in an area of focal injury in the immature central nervous system. Although we have not yet attempted to produce electroshock lesions in older animals, it would be interesting to learn if such injuries would heal, as in the infant rats, with the formation of a band of myelinated fibers rather than a glial scar.

Beyond their possible employment in the studies mentioned above, these simply produced, small, cortical lesions may also serve as useful models for the study of alterations in function after minimal brain damage sustained in infancy. Preliminary work suggests, for example, that the threshold for electrically-induced maximal seizures (criterion: tonic extension of the hind-limbs), examined at 3 to 4 months of age, is lower in such brain-injured animals than in sham-treated controls. As might be expected, the incidence of sound-induced seizures, studied at 15 to 19 months, is not altered by the electroshock lesions. Most interesting would be a determination of the functional activity and connections of the nerve fibers newly populating the originally injured area. Some electrophysiologic work related to this question has been done in the rat (Kruger & Clemente, 1964) and cat (Malis, 1964) with laminar cortical lesions, similar to ours, produced by heavy ionized particles from a linear accelerator or cyclotron beam. The special characteristics of the electroshock (and the heavy particle-produced) lesions, for example, the absence of significant vascular disturbance and of glial scarring, and the repopulation of the injured area by axons which

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myelinate, and possibly by dendrites, suggest that they may be particularly useful for developmental correlative studies, involving fine behavioral determinations, electrophysiology, and morphology, of some aspects of cortical function.

Notes

This paper provides the original and full description of the method for injuring the brain used in a study published earlier this year (Cowen, Geller, & Wolf, 1970).

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